Akshay K. Jagadish

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Position

Peretsman Scully Postdoctoral Fellow — *Princeton, USA*
Princeton University — *2025 - ongoing*
Independent researcher in the Natural and Artificial Minds program at the Princeton AI Lab

Education

Ph.D. in Computer Science — *Tübingen, Germany*
University of Tübingen · Max Planck Institute for Biological Cybernetics · Helmholtz Munich — *2021 - 2025*
Meta-Learning: A Unifying Framework for Testing Theories of Human Learning

M.Sc. in Computational Neuroscience — *Tübingen, Germany*
University of Tübingen · Max Planck Institute for Biological Cybernetics — *2018 - 2020*
Compositional Reinforcement Learning in Minds and Machines

B.Tech. in Electrical and Electronics Engineering — *Surathkal, India*
National Institute of Technology Karnataka · Ecole Polytechnique Federale de Lausanne — *2013 - 2017*
Structural and Functional Correlates of Personality

Experience

Doctoral Thesis — *Tübingen, Germany*
University of Tübingen · Max Planck Institute for Biological Cybernetics · Helmholtz Munich — *Apr. 2021 - Aug. 2025*
Computational Principles of Intelligence Lab · Advisors: Dr. Marcel Binz and Prof. Eric Schulz

Master Thesis — *Tübingen, Germany*
Max Planck Institute for Biological Cybernetics — *Jun. 2020 - Dec. 2020*
Computational Principles of Intelligence Lab · Advisors: Dr. Marcel Binz and Prof. Eric Schulz

Graduate Research Assistant — *Tübingen, Germany*
University of Tübingen — *Nov. 2018 - Mar. 2021*
Sinz Lab · Advisors: Prof. Fabian Sinz and Prof. Edgar Walker

Graduate Research Assistant — *Tübingen, Germany*
Max Planck Institute for Biological Cybernetics — *Nov. 2019 - Feb. 2020*
Computational Neuroscience Lab · Advisor: Prof. Peter Dayan

AI Researcher — *Mumbai, India*
Wadhwani Institute for Artificial Intelligence — *May 2018 - Sep. 2018*
AI for Social Impact · Advisor: Dr. Rahul Panicker

Postbaccalaureate Research Assistant — *Minnesota, USA*
University of Minnesota, Twin-cities — *Jul. 2017 - Feb. 2018*
Computational Visual Neuroscience Lab · Advisor: Prof. Kendrick Kay

Undergraduate Thesis — *Lausanne, Switzerland*
Ecole Polytechnique Federale de Lausanne — *Aug. 2016 - May 2017*
Medical Image Processing Lab · Advisors: Prof. D. van de Ville and Prof. P. Giannakopoulos

Undergraduate Research Assistant — *Bangalore, India*
Indian Institute of Science — *May 2015 - May 2017*
Computational Tomography Lab · Advisor: Prof. Kasi Rajgopal

Honors, Awards, & Fellowships

2026 **Top Reviewer**, Top 10 % reviewer (main track) for International Conference for Learning Representations *USA*
2026 **NatGeo Storyteller**, Top 12 members selected to participate in National Geography Story Tellers Workshop *USA*
2025 **Top Reviewer**, Top 10 % reviewer (main track) for Neural Information Processing Systems (NeurIPS) *USA*
2025 **Natural and Artificial Mind Postdoctoral Fellowship**, Top 5 candidates given a fellowship from the Scully Peretsman Foundation to conduct independent research at the Princeton AI Lab *USA*
2025 **Ph.D. with Magna cum Laude**, for exceptional research conducted during doctoral studies *Germany*
2024 **Top Reviewer**, Top 10 % reviewer (main track) for Neural Information Processing Systems (NeurIPS) *Canada*
2024 **ELLIS Winter School on Foundation Models**, Top 40 students selected from 607 to present their research *Netherlands*
2023 **Analytical Connectionism**, Top 35 students selected to attend the summer course at the Gatsby Computational Neuroscience Unit in University College London *UK*
2020 **SMARTSTART Fellowship in Computational Neuroscience**, Top 15 students awarded a travel budget of 1000 euros and mentorship from Prof. Peter Dayan and Prof. Fabian Sinz *Germany*
2019 **Dean's List**, Top 3 performers in the summer semester for M.Sc. program in Computational Neuroscience *Germany*
2018 **Dean's List**, Top 3 performers in the winter semester for M.Sc. program in Computational Neuroscience *Germany*
2018 **Max Planck Society Scholarship**, Top 5 students selected to undertake M.Sc in Computational Neuroscience at the University of Tübingen *Germany*
2017 **Harvard Young Scientist Development Program**, Top 25 students selected for training in neuroscience *USA | India*
2016 **Summer Research Program**, Top 20 students funded to conduct research at EPFL *Switzerland*
2016 **Summer Research Fellowship Program**, Top 10 % students funded to conduct research at the Indian Institute of Science *India*
2013 **Ranked Top 0.1%**, Karnataka Common Entrance Test among 150,000 students *India*
2013 **Ranked Top 0.1%**, COMED-K among 200,000 students *India*
2011 **Most Consistent Performer of the Batch**, High school at Presidency School *India*

Press

2025 **New York Times**, An opinion piece called *"Scientists Use A.I. to Mimic the Mind, Warts and All"* on our article A foundation model to predict and capture human cognition [URL] *AoE*
2025 **New York Times**, A science-education piece called *"Digital Therapists Get Stressed Too, Study Finds"* on our article "Chat-GPT on the Couch": Assessing and Alleviating State Anxiety in Large Language Models [URL] *AoE*
2025 **Fortune Magazine**, Interview to discuss our article "Chat-GPT on the Couch": Assessing and Alleviating State Anxiety in Large Language Models [URL] *AoE*
2025 **ScienceDaily**, An accessible take on our article "Chat-GPT on the Couch": Assessing and Alleviating State Anxiety in Large Language Models [URL] *AoE*
2024 **Tagesspiegel**, An accessible article in German called *"How does ChatGPT "think"?: A chatbot goes to a psychologist"* on our article "Chat-GPT on the Couch": Assessing and Alleviating State Anxiety in Large Language Models [URL] *AoE*

Service

2025- **Reviewer**, International Conference on Machine Learning (ICML) *AoE*
2024- **Reviewer**, Neural Information Processing Systems (NeurIPS) *AoE*
2023- **Reviewer**, International Conference on Learning Representations (ICLR) *AoE*
2023- **Reviewer**, Cognitive Computational Neuroscience (CCN) *AoE*
2022- **Reviewer**, Annual Meeting of the Cognitive Science Society (CogSci) *AoE*

Teaching

2023 **Teaching Assistant**, "Computational Cognitive Science" course at the Graduate Training Center for Neuroscience, University of Tübingen *Germany*
2022 **Tutor**, Tutorial on "Meta-Reinforcement Learning" at the Max Planck Institute for Biological Cybernetics *Germany*

Supervision

2026 **Lan Pan**, Research project on "PSYCHE agent to interact with large-scale psychology human dataset" at the University of Osnabrück. *USA*

2026 **Hanbo Xie**, Doctoral research project on "Think-Aloud Reshapes Automated Cognitive Model Discovery Beyond Behavior" at the Georgia Institute of Technology. Converted to a publication at the Computational Cognitive Neuroscience Conference [PDF] *USA*

2026 **Jerry Han**, Bachelor research project on universal preconditioning aids human function learning at the Princeton University *USA*

2026 **Alex Kawaja**, Bachelor research project on universal preconditioning aids human function learning at the Princeton University *USA*

2025 **Liyi Zhang**, Doctoral research project on "Distilling inductive reasoning into LLMs" at the Princeton University. Preprint out on Arxiv [PDF] *USA*

2025 **Solim Legris**, Doctoral research project on "Neuro-symbolic models for ARC benchmark" at the Princeton University. *USA*

2025 **Younes Strittmatter**, Doctoral research project on "Pipeline for Systematic Curation of Human Experiments" at the Princeton University. Converted to an abstract in the Cognitive Science Conference [PDF] *USA*

2025 **Daniel Braga**, Master research project on "Automated representational structure discovery" at the Princeton University. Converted to a publication in the Cognitive Science Conference [PDF] *USA*

2024 **Elif Kara**, Master research project on "Human decision-making in the wild" at the University of Münich *Germany*

2023 **Johannes Schubert**, Master thesis on "Rational Analysis of Optimism Bias" at the University of Tübingen. Converted to a publication at the *International Conference on Machine Learning (ICML)* [PDF] *Germany*

Invited talks

2026 **Cognitive Science Conference (CogSci)**, Workshop on "Laying the foundations for foundation models of cognition" *Brazil*

2026 **Massachusetts Institute of Technology**, Weekly Seminar Series at Computational Human Intelligence Focused Research Organization (CHI-FRO) *USA*

2026 **Indiana University**, Workshop on "Defining the Formal Foundations of Intelligences" at the Center for Possible Minds *USA*

2026 **Diverse Intelligence Summit**, Workshop on "AI for Scientific Discovery" *USA*

2026 **Computational and Systems Neuroscience (COSYNE)**, Workshop on "AI for Interpretable Model Discovery in Neuroscience" *Portugal*

2025 **Cornell-CUNY-UC Davis**, Cross-lab meeting part of an NSF grant on "Collective behavior in smart environments" *USA*

2025 **New York University**, NYUConcats talk series – the longest running scientific discussion groups in the Department of Psychology at NYU *USA*

2025 **Princeton University**, Natural and Artificial Minds monthly talk series in the the Princeton AI Lab *USA*

2025 **Princeton University**, Workshop on Automated Discovery of Mind and Brain *USA*

2025 **University of Osnabrück**, Lab meeting at the Laboratory for Automated Scientific Discovery of Mind and Brain *Germany*

2024 **Cognitive Science Conference (CogSci)**, Workshop on "In-context learning in natural and artificial intelligence" [URL] *Netherlands*

2024 **Cognitive Science Conference (CogSci)**, Workshop on "Compositionality in minds, brains and machines: A unifying goal that cuts across cognitive sciences" [URL] *Netherlands*

2024 **Indian Institute of Science**, Seminar talk at the Center for Neuroscience *India*

2024 **Princeton University**, Lab meeting at Computational Cognitive Science Lab *USA*

2023 **Helmholtz Münich**, Joint lab retreat with Explainable Machine Learning Lab *Germany*

2023 **University of Oxford**, Lab meeting at the Human Information Processing Lab *United Kingdom*

2022 **Max Planck Institute for Human Cognitive and Brain Sciences**, Joint lab retreat with Doeller Lab *Germany*

2021 **Stanford University**, Joint lab retreat with Human Information Processing and Causality in Cognition Lab *USA*

2019 **University of Tübingen**, Workshop on "Causality in Neuroscience" at Neuroscience Conference for Young Scientists [URL] *Germany*

Organization

2026	**Organizer**, Workshop on “AI for scientific discovery” at the Diverse Intelligences workshop series	*Princeton*
2024	**Co-organizer**, “Connecting Minds and Machines a Foundation Model Approach to Learning” symposium at Helmholtz Pioneer Campus, Munich	*Germany*
2024	**Co-organizer**, “In-context learning in natural and artificial intelligence” workshop at CogSci-2024	*Netherlands*
2023	**Co-organizer**, Laboratory Retreat of Computational Principles of Intelligence Lab	*Germany*
2022	**Co-organizer**, Computation and Cognitive Tübingen Summer School (CaCTüS) aimed specifically at young scientists held back by personal, financial, regional or societal constraints.	*Germany*
2019	**Co-organizer**, “Causality in Neuroscience” Workshop at Neuroscience Conference for Young Scientists	*Germany*
2020	**Volunteer**, Machine Learning Summer School (MLSS) held at MPI for Intelligent Systems, Tübingen	*Germany*

Publications

* equal contribution, # alphabetical ordering

Jagadish, A. K.*, Strittmatter, Y.*, Jacoby, N., Kachergis, G., Schulz, E., Daw, N., Chandramouli, S. H., & Griffiths, T. (2026). Closing the Loop to Discover Psychological Theories with an Automated Cognitive Scientist. Preprint [PDF]

Zhang, L., **Jagadish, A. K.**, Lake, B., & Griffiths, T. (2026). Using Probabilistic Programs to Train Inductive Reasoning in Large Language Models. Preprint [PDF]

Binz, M., ..., **Jagadish, A. K.**#,..., & Schulz, E. (2026). Post-training makes large language models less human-like. Preprint [PDF]

Chandramouli, S., Kachergis, G., **Jagadish, A. K.** (2026). Automated Adversarial Collaboration for Advancing Theory Building in the Cognitive Sciences. In the 9th Annual Conference Cognitive Computational Neuroscience [PDF]

Xie, H., **Jagadish, A. K.**, Pan, L., Wilson, R. C. (2026). Think-Aloud Reshapes Automated Cognitive Model Discovery Beyond Behavior. In the 9th Annual Conference Cognitive Computational Neuroscience [PDF]

Jagadish, A. K., Witte, K., Mathony, M., Binz, M., & Schulz, E. (2026). Can we automatize scientific discovery in the cognitive sciences? Preprint [PDF]

Braga, D., Marjieh, R.*, **Jagadish, A. K.***, Sucholutsky, I., Jacoby, N., & Griffiths, T. L. (2026). Automated Discovery of Psychological Representations using Language Model Agents. In the Proceedings of the Annual Meeting of the Cognitive Science Society [PDF]

Jagadish, A. K.*, Rmus, M.*, Mathony, M., & Schulz, E. (2026). Modeling individual differences in learning and memory using large language models. In the Proceedings of the Annual Meeting of the Cognitive Science Society [PDF]

Strittmatter, Y., Griffiths, T. L., & **Jagadish, A. K.** (2026). Pipeline for systematic curation of human experiments. In the Proceedings of the Annual Meeting of the Cognitive Science Society [PDF]

Jagadish, A. K., Thalmann, M., Coda-Forno, J., Binz, M. & Schulz, E. (2025). Meta-learning ecological priors from large language models captures human learning and decision making. Preprint [PDF]

Rmus, M.*, **Jagadish, A. K.***, Mathony, M., & Schulz, E. (2025). Generating computational cognitive models using large language models. In the 39th Annual Conference on Neural Information Processing Systems (NeurIPS) [PDF]

Binz, M., **Jagadish, A. K.**, Rmus, M., & Schulz, E. (2025). Automated scientific minimization of regret. In the AI4Science Workshop at the 39th Annual Conference on Neural Information Processing Systems (NeurIPS) [PDF]

Cohen, Z.*, **Jagadish, A. K.***, Hosseini, E.*, & Eckstein, M. (2025). Reinforcement learning: computational modeling of learning and decision-making Lecture Notes of the Analytical Connectionism Summer School 2023 and 2024. Proceedings of Machine Learning Research (PMLR) [PDF]

Jagadish, A. K., Thalmann, M., Coda-Forno, J., Binz, M. & Schulz, E. Bounded ecologically rational meta-learned inference explains human category learning. (2025). Proceedings of the Annual Meeting of the Cognitive Science Society [PDF]

Binz, M., ..., **Jagadish, A. K.**#,..., & Schulz, E. (2025). Centaur: a foundation model of human cognition. Nature [PDF]

Demircan, C.*, Saanum, T.*, **Jagadish, A. K.**, Binz, M., & Schulz, E. (2025). Sparse autoencoders reveal temporal difference learning in large language models. In the 13th International Conference on Learning Representations (ICLR) [PDF]

Jagadish, A. K., Coda-Forno, J., Thalmann, M., Schulz, E., & Binz, M. (2024). Human-like category learning by injecting ecological priors from large language models into neural networks. In the 41st International Conference on Machine Learning (ICML) [PDF]

Schubert, J., **Jagadish, A. K.**, Binz, M., & Schulz, E. (2024). In-context learning agents are asymmetric belief updaters. Proceedings of the 41st International Conference on Machine Learning (ICML) [PDF]

Jagadish, A. K., Binz, M., Saanum, T., Wang, J. X., & Schulz, E. (2024). Zero-shot compositional reasoning in a reinforcement learning setting. Preprint [PDF]

Ben-Zion, Z., Witte, K.*, **Jagadish, A. K.***, Duek, O., Harpaz-Rotem, I., Khorsandian, M., ... & Spiller, T. R. (2024). "Chat-GPT on the couch": assessing and alleviating state anxiety in large language models. npj Digital Medicine [PDF]

Coda-Forno, J.*, Witte, K.*, **Jagadish, A. K.**, Binz, M., Akata, Z., & Schulz, E. (2023). Inducing anxiety in large language models increases exploration and bias. Preprint [PDF]

Binz, M., Dasgupta, I., **Jagadish, A. K.**, Botvinick, M., Wang, J.X., & Schulz, E. (2023). Meta-learned models of cognition. Behavioral and Brain Sciences [PDF]

Schubert, J., **Jagadish, A. K.**, Binz, M., & Schulz, E. (2023). A rational analysis of optimism bias using meta-reinforcement learning. In the 6th Annual Conference on Cognitive Computational Neuroscience (CCN) [PDF]

Jagadish, A. K., Saanum, T., Wang, J. X., Binz, M., & Schulz, E. (2022). Probing compositional inference in natural and artificial agents. In 5th Multidisciplinary Conference on Reinforcement Learning and Decision Making (RLDM)

Bashiri, M.*, Walker, E.*, Lurz, K. K., **Jagadish, A. K.**, Muhammad, T., Ding, Z., ... & Sinz, F. (2021). A flow-based latent state generative model of neural population responses to natural images. In Advances in Neural Information Processing Systems (NeurIPS) [PDF]

Lurz, K. K., Bashiri, M., Willeke, K. F., **Jagadish, A. K.**, Wang, E., Walker, E. Y., ... & Sinz, F. (2021). Generalization in data-driven models of primary visual cortex. In International Conference on Learning Representations (ICLR) [PDF]

Rodriguez, C.*, **Jagadish, A. K.***, Meskaldji, D. E., Haller, S., Herrmann, F., Van De Ville, D., & Giannakopoulos, P. (2019). Structural correlates of personality dimensions in healthy aging and MCI. Frontiers in psychology [PDF]

Jagadish, A. K., Goswami, S., Saha, P., Chakrabarty, S., & Rajgopal, K. (2016). Artificial Bee Colony (ABC) based variable density sampling scheme for CS-MRI. In 2016 IEEE Region 10 Conference (TENCON) [PDF]